

Rollback-consistent state transactions and auditable verification of an Abaqus multiphase poromechanics user element

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Abstract

Nonlinear multiphysics user elements combine history-dependent variables, branch-dependent constitutive responses, and solver-driven trial evaluations. A converged endpoint alone therefore cannot reveal whether rejected calls contaminate physical history, whether a returned matrix differentiates the implemented residual, or whether numerical representation changes an audit result. This study presents an auditable verification framework for one frozen equal-order Q4 displacement–pressure–saturation Abaqus UEL, without proposing a new constitutive model. Six falsifiable conditions separate committed physical history, trial candidates, algorithmic quantities, and diagnostics. Source tracing and equal-state replay expose a capillary-reference cache whose value depends on rejected call history; moving that quantity to solver-managed state removes the identified path. A bidirectional ledger then links three residuals, nine matrix blocks, 48 state slots, units, and active branches. Lifecycle replay, blockwise and mixed directional differentiation, stable-active-set analysis, assembled-patch checks, and independent repetition are evaluated as distinct evidence layers. The strict binary64 mixed-direction record remains 36/39, while the three lower-clip cases are supported separately by an active-set oracle and binary128 differences. The predefined Newton-order criterion and a common local basin were not established, and

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those outcomes are retained rather than replaced by favorable subsets. The framework provides bounded evidence for rollback-consistent state ownership, traceability, and selected branch derivatives in the tested domain. Engineering qualification remains separate: equal-order stability, native-balance closure, mesh and time convergence, full-domain validity, and production readiness are not established.

Keywords: rollback consistency, auditable verification, user-defined finite element, multiphase poromechanics, Jacobian verification, numerical reproducibility

1. Introduction

1.1. Reliability challenges in nonlinear multiphysics user elements

Nonlinear multiphase poromechanics couples deformation, pore pressure, saturation-dependent transport, and history-dependent constitutive response across several numerical layers. Its continuum basis extends classical consolidation and poromechanics to partially saturated and multiphase settings (Biot, 1941; Coussy, 2003; Borja and Alarcón, 1995; Borja, 2004). A nominal time increment can include repeated Newton iterations, line searches, failed attempts, terminal calls, and cutbacks before the host solver accepts a new state. Calls made during this process are alternative trials originating from one committed beginning-of-increment state, rather than independent evaluations of unrelated configurations. Numerical credibility therefore depends on whether the user implementation obeys the host transaction as well as on whether it represents the intended equations.

Classical constitutive algorithms distinguish the converged beginning state from an endpoint candidate. Ortiz and Popov separated integration accuracy, incremental consistency, and nonlinear stability for local updates (Ortiz and Popov, 1985). Simo and Taylor showed that every return-mapping trial should be reconstructed from the converged beginning state and that the tangent must differentiate the discrete response used by the residual if the asymptotic Newton rate is to be retained (Simo and Taylor, 1985). Consistent-tangent construction and numerical differentiation of algorithmic constitutive updates were subsequently systematized for computational inelasticity (Miehe, 1996; Simo and Hughes, 1998; de Souza Neto et al., 2008). These results establish deterministic state transitions and residual-tangent compatibility. They do

not by themselves prescribe rollback of physical memory privately owned by a compiled user extension.

That distinction is consequential. Solver-managed state variables can be restored after a rejected attempt, whereas language-level static arrays, module variables, common blocks, or file-backed caches can survive rejection. If such persistence subsequently enters a residual or tangent, two calls with identical declared inputs can produce different outputs solely because they were preceded by different rejected histories. An implementation can remain finite and advance for many increments while its effective nonlinear residual is no longer a function of the declared state alone. This defect is categorically different from a declared quasi-Newton approximation: an approximate tangent changes a solution operator while preserving the residual, whereas hidden physical-state mutation changes the residual itself.

1.2. Why conventional verification is insufficient

Conventional implementation checks often emphasize successful termination, final displacement or pressure fields, and a small terminal residual. Those observations are necessary but do not determine whether a rejected trial modified hidden state, whether every active branch uses the derivative implied by its value path, whether a raw returned matrix contains nonphysical numerical terms, or whether a benchmark failure belongs to the implementation, the algebraic solver, or the reference itself. These questions require separate lifecycle, traceability, derivative, assembly, and benchmark-qualification evidence.

Poromechanical coupling algorithms create a second and independent approximation layer. Reservoir-geomechanics studies have long compared fully coupled and sequential solution strategies (Settari and Mounts, 1998; Settari and Walters, 2001; Dean et al., 2006). Fixed-strain, fixed-stress, and related partitioned methods replace a monolithic solve with declared block iterations (Kim et al., 2011; Castelletto et al., 2013; Mikelić and Wheeler, 2013; White et al., 2016). Their convergence, accuracy, optimization, multi-scale extension, and parameter robustness depend on the stated model and splitting assumptions (Castelletto et al., 2015; Dana et al., 2018; Storvik et al., 2019; Hong et al., 2020). Block-factorization analyses relate such schemes to approximations of monolithic Schur complements (White et al., 2016). For nonlinear unsaturated or multiphase systems, L-schemes, modified Picard, Newton, and accelerated fixed-stress variants exchange local speed for robustness (Both et al., 2019). A single flow solve followed by a single mechanics

63 solve is not equivalent to a converged two-way fixed-stress iteration.

64 The pressure variable transmitting multiphase effects to mechanics is
65 another independent choice. Effective-stress and stress-saturation frameworks
66 for unsaturated soils are model dependent and are tied to their constitutive
67 assumptions (Nuth and Laloui, 2008; Coussy et al., 2010; Sheng et al., 2008;
68 Zhou et al., 2012; Sun et al., 2010). Strong-capillarity analyses likewise
69 demonstrate that an energy-compatible equivalent pore pressure can influence
70 coupled stability (Kim et al., 2013). These observations motivate, but do
71 not uniquely select, the empirical weighting used by the present element. A
72 defensible implementation must trace the declared pressure definition through
73 the mechanics residual and every coupling derivative. Similarly, convergence
74 of a fixed-stress outer iteration does not replace a spatial pressure-mode audit
75 (Aronson et al., 2024), and nested fixed-stress schemes with explicit inner and
76 outer states do not automatically solve host rollback of user-owned physical
77 memory (Correa and Murad, 2025).

78 *1.3. Objective and bounded contributions*

79 The relevant gap is therefore cross-layer rather than constitutive novelty.
80 Official Abaqus documentation defines the UEL arrays and identifies **SVARS** as
81 solver-managed state for general nonlinear steps (Dassault Systèmes SIMULIA,
82 2026), while classical consistent-tangent literature requires the matrix to
83 differentiate the discrete response (Simo and Taylor, 1985; Miehe, 1996).
84 Neither statement alone verifies that private user memory is rollback safe,
85 that every returned block differentiates the implemented residual, or that a
86 locally accurate derivative yields a useful nonlinear basin. The objective is
87 consequently not to certify a generally deployable solver, but to establish an
88 auditable framework that can expose and bound these distinct implementation
89 risks for one frozen UEL.

90 Four bounded contributions are made. First, a rollback-consistent transac-
91 tion protocol uses six falsifiable conditions to separate committed, trial, algo-
92 rithmic, and diagnostic state. Second, source-extracted replay demonstrates
93 rejected-history contamination by a capillary-reference cache and defines a
94 solver-managed repair pattern; repaired-build Abaqus patch tests provide
95 a separate, narrower host check. Third, an equation-to-code-state-Jacobian
96 ledger links the weak equations, source routines, nine matrix blocks, 48 state
97 slots, units, and branch rules. Fourth, an evidence-bounded workflow sepa-
98 rates lifecycle replay, directional and active-set differentiation, assembled patch
99 checks, Newton order, and local-basin geometry while retaining negative and

100 inconclusive outcomes. The study is an auditable implementation-verification
 101 case, not a validation of a new geotechnical constitutive law.

102 The scope is correspondingly narrow. The corrected physical derivative is
 103 supported only over registered smooth and stable-active-set P1 states. Equal-
 104 order field stability, native balance closure, mesh and time-step convergence,
 105 a documented engineering benchmark, a common Newton basin, and full-
 106 domain performance remain unestablished. These omissions are explicit
 107 resubmission gates rather than implied achievements.

108 2. Multiphase poromechanics UEL formulation

109 2.1. Primary fields and phase variables

110 The four-node plane-strain element uses the primary fields

$$\mathbf{x} = \{\mathbf{u}, p_w, S_n\}, \quad (1)$$

111 where $\mathbf{u}$ is solid displacement, p_w is wetting-phase pressure, and S_n is
 112 non-wetting-phase saturation. The wetting saturation is $S_w = 1 - S_n$. With
 113 residual saturations S_{wr} and S_{nr} , the effective wetting saturation is evaluated
 114 as

$$S_e = \text{clip} \left(\frac{S_w - S_{wr}}{1 - S_{wr} - S_{nr}}; \varepsilon, 1 - \varepsilon \right), \quad (2)$$

115 The clipping operation is part of the declared numerical closure. In its
 116 smooth interior, $\partial S_e / \partial S_n = -1 / (1 - S_{wr} - S_{nr})$; on an active endpoint clip,
 117 the derivative of the clipped argument is zero. Exact switching points are
 118 treated as nonsmooth and are not included in smooth central-difference claims.

119 The capillary and relative-permeability paths use the implemented Brooks-
 120 Corey-type power laws (Brooks and Corey, 1964),

$$p_c = p_b (S_e^{-1/\lambda_{BC}} - 1), \quad k_{rw} = S_e^{n_w}, \quad k_{rn} = (1 - S_e)^{n_n}. \quad (3)$$

121 Effective-saturation descriptions of retention and hydraulic conductivity
 122 have several established forms, including the widely used closed-form model
 123 of van Genuchten (1980). The power laws above are the frozen engineering
 124 choices of the audited element rather than a calibration claim for that or any
 125 other retention family. Analytical derivatives are evaluated on the same active
 126 branch as the corresponding values. The present study reports these relations

127 exactly as implemented and does not claim that their parameterization or
 128 calibration range transfers beyond the frozen benchmark.

129 2.2. Equivalent pressure and mechanical coupling

130 The implementation defines

$$\chi(S_e) = 1 - S_e, \quad p_{eq} = p_w + \chi(S_e)p_c. \quad (4)$$

131 Because $p_n = p_w + p_c$, this can also be written as $p_{eq} = S_e p_w + (1 - S_e)p_n$.
 132 On a smooth active branch,

$$\frac{\partial p_{eq}}{\partial p_w} = 1, \quad \frac{\partial p_{eq}}{\partial S_n} = \frac{\partial \chi}{\partial S_n} p_c + \chi \frac{\partial p_c}{\partial S_n}. \quad (5)$$

133 The value and first derivatives follow one branch-aware code path. An
 134 independent audit found implementation-level derivative agreement, but
 135 this establishes algebraic consistency only. The relation is retained as a
 136 Bishop-type engineering closure and is not substituted for thermodynamically
 137 structured or effective-stress formulations in the literature (Nuth and Laloui,
 138 2008; Coussy et al., 2010; Borja, 2004). The present evidence does not provide
 139 a project-specific Helmholtz free energy, unique work-conjugate pressure, or
 140 dissipation proof; rollback-safe ownership therefore must not be conflated
 141 with thermodynamic consistency.

142 The frozen mechanics model is isotropic small-strain plane strain with
 143 engineering shear strain. With tensile stress positive,

$$\boldsymbol{\sigma}' = \mathbf{D}\boldsymbol{\epsilon}, \quad \mathbf{D} = \begin{bmatrix} \lambda + 2G & \lambda & 0 \\ \lambda & \lambda + 2G & 0 \\ 0 & 0 & G \end{bmatrix}, \quad (6)$$

144 where $\lambda = E\nu/[(1 + \nu)(1 - 2\nu)]$ and $G = E/[2(1 + \nu)]$. The implemented
 145 mechanics residual is

$$\mathbf{R}_u = \int_{\Omega_e} \mathbf{B}^T \boldsymbol{\sigma}' d\Omega - \int_{\Omega_e} \mathbf{B}^T \alpha p_{eq} \mathbf{m} d\Omega, \quad \mathbf{m} = (1, 1, 0)^T. \quad (7)$$

146 Thus $K_{uu} = \int B^T D B d\Omega$, while K_{up} and K_{us} differentiate the unchanged
 147 equivalent-pressure force with respect to p_w and S_n . These are code-level
 148 definitions of the audited element, not a calibration or validation of the
 149 equivalent-stress closure.

150 *2.3. Pressure and saturation residuals*

151 With N_i denoting a Q4 test function, $\mathbf{m} = (1, 1, 0)^T$, and superscripts
 152 k and n denoting the current trial and beginning-of-increment committed
 153 states, respectively, the implemented pressure residual is

$$R_{p,i} = \int_{\Omega_e} \left[N_i c_{p,\text{impl}} \frac{p_w^k - p_w^n}{\Delta t} + N_i \alpha \frac{\varepsilon_v^k - \varepsilon_v^n}{\Delta t} + \nabla N_i \cdot (\lambda_w \nabla p_w^k) \right] d\Omega, \quad (8)$$

154 where $c_{p,\text{impl}} = \text{PROPS}(4)$ and $\lambda_w = K k_{rw} / \mu_w$. The legacy source annota-
 155 tion for **PROPS(4)** is inconsistent with its dimensional role in the discrete equa-
 156 tion; the paper therefore reports it only as the implemented pressure-storage
 157 coefficient and assigns no independent calibrated-parameter provenance. The
 158 pressure assembler contains no explicit well or boundary source term; such
 159 loading must be traced to the input deck rather than attributed to the UEL
 160 residual.

161 The saturation assembler optionally evaluates a guarded trial value

$$S_n^{\text{use}} = \text{clip}_{[0,1]} [S_n^n + \beta_S (S_n^k - S_n^n)] \quad (9)$$

162 and assembles

$$R_{s,i} = \int_{\Omega_e} \left[N_i \phi \rho_n \frac{S_n^{\text{use}} - S_n^n}{\Delta t} + N_i \rho_n \beta_\phi S_n^{\text{use}} \frac{\varepsilon_v^k - \varepsilon_v^n}{\Delta t} \right. \\ \left. + \nabla N_i \cdot \{ \lambda_n [\nabla p_w^k + p'_c(S_n^k) \nabla S_n^k] \} \right] d\Omega, \quad (10)$$

163 where

$$\lambda_n = \frac{\rho_n K k_{rn}}{\mu_n}, \quad \frac{\partial \lambda_n}{\partial S_n} = \frac{\rho_n K}{\mu_n} \frac{\partial k_{rn}}{\partial S_n}. \quad (11)$$

164 The storage/strain path and the closure/flux path deliberately retain
 165 their distinct saturation values. Linearization must therefore differentiate
 166 each declared path rather than merge them into a numerically convenient
 167 surrogate.

168 *2.4. Stress-dependent permeability, parameters, and numerical controls*

169 The scalar absolute permeability uses the mean in-plane effective stress
 170 $\sigma'_m = (\sigma'_{xx} + \sigma'_{yy})/2$ and is evaluated as

$$K_{raw} = K_0 \exp[a_\sigma(\sigma'_m - \sigma'_0)] s_{cb}, \quad (12)$$

$$s_{cb} = 1 + a_{cb}(CB_{GP} - CB_0), \quad (13)$$

$$K = \text{clip}(K_{raw}; K_{min}, K_{max}). \quad (14)$$

171 The registered profile sets $a_{cb} = 0$, so $s_{cb} = 1$. On the interior exponential
 172 branch, $\partial K / \partial \sigma'_m = a_\sigma K_{raw}$; on an active permeability bound the derivative
 173 of the clipped value is zero, and exact switching points are excluded. The
 174 value is scalar and isotropic; no permeability tensor or anisotropy claim is
 175 made.

176 Table 1 reports every input used by the tested single-element harness.
 177 Parameters 23–27 are disabled diagnostic or freeze switches and are listed in
 178 Supplementary Table S1. For the present paper, **PROPS(4)** is defined uniquely
 179 by its implemented residual and tangent role as an implementation-level
 180 pressure-storage coefficient $c_{p,\text{impl}}$ with unit Pa^{-1} . The legacy source annota-
 181 tion “dimensionless” is treated as incorrect metadata, not as an alternative
 182 definition. No calibration source, constitutive identification, or sensitivity
 183 study is available, so the coefficient is not presented as a validated material
 184 property.

Table 1: Registered P1 input parameters. A dash denotes dimensionless.

No.	Symbol or role	Value	Unit	Evidence boundary
1	E	2.50×10^{10}	Pa	isotropic plane-strain elasticity
2	ν	0.300	–	isotropic plane-strain elasticity
3	α	1.00	–	equivalent-pressure coupling
4	$c_{p,\text{impl}}$ STORW	/ 1.00×10^{-6}	Pa^{-1}	implementation-level pressure-storage coefficient; calibration and sensitivity pending

No.	Symbol or role	Value	Unit	Evidence boundary
5	μ_w	1.00×10^{-3}	Pa s	wetting mobility
6	ρ_n	700	kg m ⁻³	non-wetting mass balance
7	μ_n	1.00×10^{-3}	Pa s	non-wetting mobility
8	S_{wr}	0.200	–	residual wetting saturation
9	S_{nr}	0.0500	–	residual non-wetting saturation
10	p_b	1.00×10^4	Pa	entry-pressure parameter
11	λ_{BC}	2.00	–	Brooks–Corey exponent
12	n_w	2.00	–	wetting relative-permeability exponent
13	n_n	2.00	–	non-wetting relative-permeability exponent
14	β_ϕ	1.00×10^{-2}	–	strain–saturation coupling coefficient
15	debug seed	0.100	–	disabled; no physical role
16	K_0	6.00×10^{-12}	m ²	reference absolute permeability
17	σ'_0	0	Pa	reference mean effective stress
18	a_σ	0 or 1.00×10^{-8}	Pa ⁻¹	state-specific stress sensitivity
19	CB_0	0	–	disabled coupling baseline
20	a_{cb}	0	–	disabled coupling sensitivity
21	K_{min}	1.00×10^{-20}	m ²	lower permeability bound
22	K_{max}	1.00×10^{-8}	m ²	upper permeability bound

185 Material relations and phase balances belong to the frozen engineering
 186 model. Tangent scaling, matrix-only diagonal terms, limiter controls, and
 187 increment controls belong to the numerical implementation. M1 contains the

188 stress-dependent permeability value but omits its displacement derivative in
 189 the flow rows, uses a 0.75 mechanics-coupling tangent scale, and differentiates
 190 selected saturation paths incompletely. M2a changes only those registered
 191 derivatives; the residual, closure values, and physical state are parity controls.

192 3. Rollback-consistent state-ownership architecture

193 3.1. Declared state classes

194 Let t_n denote the beginning of a candidate increment and let $\mathbf{h}_n$ be the
 195 committed physical history supplied by the host. One element call evaluates

$$\mathcal{F}(\mathbf{x}^{(k)}; \mathbf{h}_n, \boldsymbol{\theta}) = \left(\mathbf{R}^{(k)}, \mathbf{J}^{(k)}, \widehat{\mathbf{h}}^{(k)}, \mathbf{d}^{(k)} \right), \quad (15)$$

196 where $\boldsymbol{\theta}$ is immutable configuration, $\widehat{\mathbf{h}}^{(k)}$ is a candidate end state, and $\mathbf{d}^{(k)}$
 197 contains physically inert diagnostics. Four state classes are distinguished.

Table 2: State classes and permitted ownership lifetimes.

Class	Meaning	Permitted lifetime
C_n	committed physical history	Solver-managed and rollback capable
T_k	current Newton, line-search, or cutback trial	Call-local or candidate solver-managed state
A_k	declared algorithmic iteration state	Explicitly versioned and reset by the algorithm
D_k	counters, logs, and diagnostics	Persistent only if no reverse path to residual, tangent, or physical history exists

198 This classification extends the classical beginning-state/candidate-state
 199 distinction by addressing memory that the host cannot automatically restore.
 200 In the documented Abaqus/Standard UEL interface, **RHS**, **AMATRX**, and **SVARS**
 201 are user-defined outputs, and in general nonlinear steps **SVARS** enters with
 202 beginning-of-increment values and may be returned as an end-of-increment
 203 candidate (Dassault Systèmes SIMULIA, 2026). A **KINC=0** call is documented
 204 as zero-increment output and cannot be used to commit physical history.

205 These interface facts motivate the ownership contract; the stronger claim that
 206 rejected call orders leave the implementation invariant is established only for
 207 the explicitly tested paths, not inferred from documentation.

208 The source-derived per-Gauss-point inventory is listed in Supplementary
 209 Table S3. Slots 1–14 are physical or closure candidates; slots 15–32 are
 210 conditionally written mechanics diagnostics; slots 33–45 are a conditionally
 211 written output mirror with provenance fields; and only slots 46–48 remain
 212 reserved. Static use-def tracing found no read from slots 15–45 into **RHS**,
 213 **AMATRX**, or slots 1–14, but these quantities are nevertheless solver-managed
 214 diagnostic state and are subject to rollback, zero-increment, and restart
 215 noninterference requirements. Private **SAVE**, **DATA**, module, common-block,
 216 or file-backed values may be diagnostic only when no reverse physical path
 217 exists.

218 3.2. Rollback-consistency conditions

219 **RC1: committed-state immutability.** A trial call may not alter $\mathbf{h}_n$
 220 before acceptance.

221 **RC2: call-history invariance.** Identical $\mathbf{x}$, $\mathbf{h}_n$, and $\boldsymbol{\theta}$ must produce
 222 equivalent physical outputs after different rejected trial histories.

223 **RC3: rejection invariance.** Rejecting $\hat{\mathbf{h}}^{(k)}$ leaves the committed state
 224 equal to $\mathbf{h}_n$.

225 **RC4: acceptance atomicity.** A candidate becomes $\mathbf{h}_{n+1}$ only through
 226 the host acceptance event; the user element does not maintain a competing
 227 private commit.

228 **RC5: terminal-call noninterference.** Zero-increment, output-only,
 229 and terminal calls may not change the next physical residual or committed
 230 history.

231 **RC6: serialization equivalence.** When restart is claimed, serialization
 232 and restoration of the committed state must reproduce the uninterrupted
 233 update within a frozen envelope.

234 Equivalence under RC2 and RC6 may be bitwise, numerical, or engineering-
 235 level, but the envelope and compared fields must be fixed before the test. A
 236 finite result is a prerequisite, not an equivalence criterion.

237 3.3. Single-source principle and persistent-state inventory

238 Every history variable that can affect a residual, tangent, or subsequent
 239 physical state must have one authoritative committed source,

$$\text{source}(h_i^n) = \text{SVARS}(s_i). \quad (16)$$

240 If a solver-managed slot and a writable static cache represent the same
 241 quantity, update order becomes hidden state and RC2 is generally lost. Per-
 242 sistent storage remains permissible for immutable configuration or diagnostics
 243 without a reverse physical path. Explicit iterative history, such as acceleration
 244 vectors, belongs to A_k and requires a declared version and reset rule.

245 The element allocates 48 state slots per Gauss point and 192 per element.
 246 Slots 1–14 contain primary, closure, stress, CAP_REF, and validity candi-
 247 dates. Slots 15–32 are populated only by the registered mechanics-debug
 248 gate, whereas slots 33–45 mirror selected fields and their source, time, and
 249 increment metadata when the output-source gate is active. Slots 46–48 are
 250 initialized to zero and otherwise unused. A source-level audit found no reverse
 251 path from slots 15–45 into the physical residual, matrix, or slots 1–14; this
 252 static result does not replace dynamic output-on/off, rejection, zero-increment,
 253 and restart checks for every flag combination. Static scanning also found no
 254 remaining active trial-updated physical history in explicit **SAVE**, **DATA**, module,
 255 or file storage after the CAP_REF repair. Absence of a known cache is not
 256 proof of all-path rollback or parallel safety.

257 3.4. CAP_REF counterexample and bounded repair

258 The audited baseline contained CAP_REF_EGP and CAP_REF_INIT as For-
 259 tran **SAVE** arrays. A trial could initialize or overwrite the capillary reference
 260 using the current $p_{eq} - p_w$, after which the saved value entered the equivalent-
 261 pressure force and mechanical residual. Abaqus could restore **SVARS** after
 262 rejection but could not restore those private arrays. The path therefore
 263 violated RC2 and RC3.

264 The repaired implementation stores CAP_REF and its valid flag in Gauss-
 265 point **SVARS**(13:14) and reads the committed input through one selector.
 266 Shared CAP_REF keys subsequently agreed exactly in the bounded lifecycle
 267 tests. This supports the statement that the identified path was made rollback-
 268 safe. It does not prove that all state slots, static objects, host terminal paths,
 269 or parallel diagnostic operations are safe.

270 4. Discrete formulation and auditable traceability

271 4.1. Element ordering and residual contract

272 The Q4 element has 16 degrees of freedom, ordered node-wise as

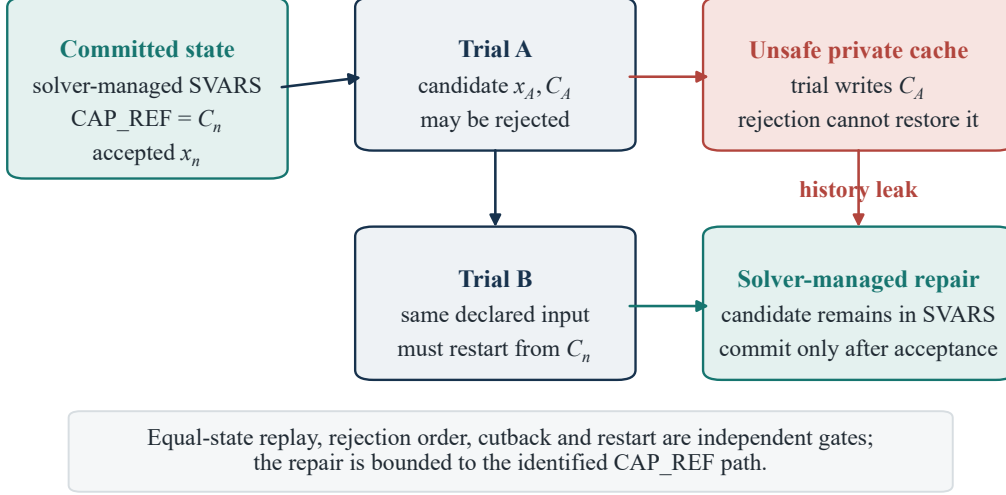


Figure 1: Host rollback restores solver-managed committed state but cannot restore a private trial-updated physical cache. Moving the identified capillary reference into solver-managed state removes that path; the diagram does not imply universal state safety.

$$\{u_x, u_y, p_w, S_n\}_a, \quad a = 1, \dots, 4. \quad (17)$$

273 The corresponding row sets are $\mathcal{I}_u = \{1, 2, 5, 6, 9, 10, 13, 14\}$,
 274 $\mathcal{I}_p = \{3, 7, 11, 15\}$, and $\mathcal{I}_s = \{4, 8, 12, 16\}$. The internal element
 275 residual is

$$\mathbf{R} = \begin{bmatrix} \mathbf{R}_u \\ \mathbf{R}_p \\ \mathbf{R}_s \end{bmatrix}, \quad \mathbf{J}_{phys} = \frac{\partial \mathbf{R}}{\partial \mathbf{x}} = \begin{bmatrix} \mathbf{K}_{uu} & \mathbf{K}_{up} & \mathbf{K}_{us} \\ \mathbf{K}_{pu} & \mathbf{K}_{pp} & \mathbf{K}_{ps} \\ \mathbf{K}_{su} & \mathbf{K}_{sp} & \mathbf{K}_{ss} \end{bmatrix}. \quad (18)$$

276 The host convention is $\text{RHS} = -\mathbf{R}$ and $\text{AMATRIX} = \mathbf{K}$. Consequently, the
 277 registered directional identity is

$$D(\text{RHS})[\mathbf{d}] \simeq -\mathbf{J}_{phys} \mathbf{d}. \quad (19)$$

278 The sign was frozen globally before directional testing; it was not selected
 279 independently for individual blocks, states, or perturbation sizes.

280 *4.2. Bidirectional traceability*

281 Traceability was established in both directions: from each equation term
 282 to its code implementation and from every returned block or state slot back to
 283 a declared equation or numerical role. The principal mapping is summarized
 284 below.

Table 3: Compact traceability overview. The auditable term–routine–block–state–unit–test ledger is Supplementary Table S2.

Equation responsibility	Residual rows	Principal derivatives	State or interface anchor
Effective-stress equilibrium and equivalent-pressure force	R_u	K_{uu}, K_{up}, K_{us}	RHS, AMATRIX, effective-stress slots, CAP_REF
Wetting storage, volumetric-strain coupling, and wetting flux	R_p	K_{pu}, K_{pp}, K_{ps}	committed p_w , ε_v , K , and closure values
Non-wetting storage, strain coupling, and pressure/capillary flux	R_s	K_{su}, K_{sp}, K_{ss}	committed S_n , branch-aware S_n^{use} , p_c , and mobility
Candidate accepted-state export	no new residual	no physical derivative	solver-managed SVARS, accepted-call provenance
Matrix stabilization and diagnostics	no physical residual unless paired	J_{num} or nonphysical output only	AMATRIX, counters, CSV/FIL writers

285 The completed RP/RS audit resolved the earlier pressure- and saturation-
 286 equation tracing gaps. It also identified where a matrix term had no paired
 287 physical residual and where a value path had been differentiated incompletely.
 288 This distinction prevents a returned nonzero entry from being treated as
 289 evidence that the corresponding equation derivative is complete.

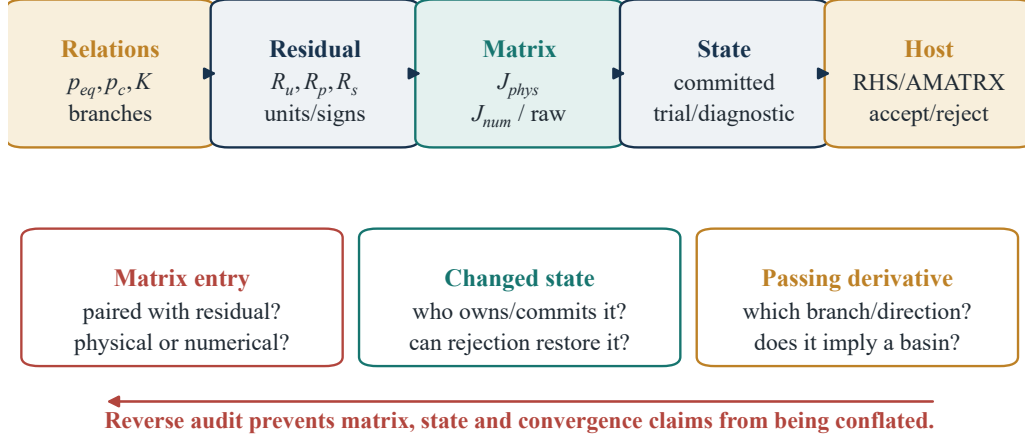


Figure 2: Bidirectional traceability links equation terms, residual rows, physical and numerical matrix blocks, state ownership and host arrays. Raw returned-matrix entries are not automatically physical derivatives.

290 4.3. Branch-aware value-and-derivative paths

291 The corrected implementation uses stateless value-and-derivative paths
 292 for stress-dependent permeability, equivalent pressure, capillary pressure,
 293 relative permeability, and the guarded saturation used by storage and strain
 294 terms. A helper returns a value, its derivative on the active branch, and
 295 a branch identifier. The residual and tangent therefore share the same
 296 clipping, limiter, and bound decisions. This fixed-active-set derivative is
 297 distinct from a generalized derivative at a switching point, consistent with
 298 the broader nonsmooth-Newton and active-set distinction (Qi and Sun, 1993;
 299 Hintermueller et al., 2002). No trial-dependent physical SAVE, DATA, module,
 300 or file cache was added.

301 For K in the interior exponential branch,

$$\frac{\partial K}{\partial \sigma'_m} = a_\sigma K, \quad \frac{\partial K}{\partial u_j} = \frac{\partial K}{\partial \sigma'_m} \frac{\partial \sigma'_m}{\partial u_j}, \quad (20)$$

302 whereas an active lower or upper permeability bound has zero derivative of
 303 the clipped value. For the saturation equation, K_{ss} differentiates the declared
 304 storage-use path, strain-saturation term, mobility, $p'_c \nabla S_n$, and the $p''_c N_j \nabla S_n$
 305 contribution. Stable active branches are tested as classical derivatives; exact

switching surfaces are retained as nonsmooth boundaries rather than assigned
a post-hoc central derivative.

4.4. Physical and numerical matrices

For audit purposes the returned matrix is decomposed as

$$\mathbf{J}_{returned} = \mathbf{J}_{phys} + \mathbf{J}_{num}, \quad \mathbf{J}_{num} = 10^{-12}\mathbf{I}_{16}. \quad (21)$$

The scalar value 10^{-12} is the legacy coefficient written to every diagonal entry in the unscaled physical-coordinate matrix. Because displacement, pressure, and saturation columns and the three residual row families have different units, this expression is not dimensionally homogeneous and is not invariant to a change of units. It is therefore neither a physical stabilization law nor evidence that the equal-order Q4 interpolation is stable. With variable and residual scaling matrices $\mathbf{X}$ and $\mathbf{W}$, its actual scaled representation is

$$\hat{\mathbf{J}}_{num} = \mathbf{W}(10^{-12}\mathbf{I}_{16})\mathbf{X}, \quad (22)$$

which is generally not a scalar identity. Directional physical-consistency tests use $\mathbf{J}_{phys} = \mathbf{J}_{returned} - \mathbf{J}_{num}$ because the guard has no paired residual term. Abaqus, however, receives $\mathbf{J}_{returned}$. Consequently, verification of $\mathbf{J}_{phys}$ cannot be promoted to a statement about the Newton behavior, unit robustness, or field stability of the host solve. Raw AMATRX is retained in parity and sensitivity records, and the failed nonlinear gates are reported separately.

5. Algorithm hierarchy

5.1. M0: legacy hidden-state baseline

M0 denotes the implementation in which the CAP_REF reference can be updated in private persistent memory during a trial. Because rejected calls can then change a later residual at identical declared inputs, M0 does not define a rollback-consistent residual map. It is retained as a falsifying counterexample, not as a production comparator.

331 5.2. M1: rollback-safe quasi-Newton baseline

332 M1 moves the identified physical reference into solver-managed committed
333 state while retaining the original mixed tangent approximations. In particular,
334 its mechanics coupling uses a 0.75 tangent scale, its stress-dependent perme-
335 ability value path lacks the corresponding displacement derivatives in flow
336 blocks, and its saturation block freezes or omits selected coefficient derivatives.
337 M1 is therefore a declared rollback-safe quasi-Newton baseline. Its residual is
338 held fixed while the mismatch of each registered block is measured.

339 5.3. M2a: branchwise physical-Jacobian candidate

340 M2a preserves the M1 physical residual and lifecycle repair while cor-
341 recting derivatives on smooth and stable active branches. It restores the
342 full equivalent-pressure mechanics derivatives, adds the stress-permeability
343 chain to K_{pu} and K_{su} , completes the declared K_{ss} branchwise derivative,
344 and separates K_{pp}^{phys} from the unpaired diagonal guard. The term physical-
345 Jacobian candidate is used deliberately: strong P1 directional evidence does
346 not establish universal Newton behavior, exact-switching differentiability,
347 full-domain validity, or equivalence of raw AMATRX and $\mathbf{J}_{phys}$.

348 5.4. Fair-comparison contract

349 Comparisons among implemented levels use identical residuals, committed
350 states, material data, time semantics, variable scaling, constraints, and branch
351 definitions. Changes in RHS, closure values, or physical candidate SVARS
352 are hard failures rather than admissible algorithmic improvements. Matrix
353 differences are accepted only when they match the preregistered derivative
354 ledger. This contract separates lifecycle repair, Jacobian correction, and
355 nonlinear-solver behavior.

356 6. Auditable verification framework

357 6.1. Orthogonal verification dimensions

358 The framework progresses from state transactions through equation-to-
359 code traceability, branchwise derivatives, assembled behavior, benchmark
360 qualification, and only then engineering deployment. Each layer asks a
361 different falsifiable question and has an independent evidence boundary.
362 Verification is therefore organized into dimensions that cannot substitute
363 for one another, following the broader distinction between code verification,

364 solution verification, validation, and numerical-error assessment (Roy, 2005;
 365 Oberkampf and Roy, 2010):

- 366 1. **Lifecycle safety:** identical declared states must be independent of
 367 rejected-call history, terminal calls, and accepted-state serialization
 368 within a frozen envelope.
- 369 2. **Residual and state parity:** a tangent-only correction must leave
 370 RHS, closure values, and physical candidate state unchanged.
- 371 3. **Directional Jacobian consistency:** block and full-matrix products
 372 are compared with directional residual derivatives over registered states,
 373 directions, and perturbation scales.
- 374 4. **Patch and assembled balance:** multi-element modes test interpola-
 375 tion, assembly, interior balance, and coupled responses.
- 376 5. **Nonlinear convergence behavior:** preregistered Newton histories,
 377 empirical order, and local-basin maps test the solver consequence of the
 378 assembled operator.
- 379 6. **Full-domain accepted-state provenance:** coverage, conservation,
 380 non-perturbation, repeatability, and parallel diagnostic safety must be
 381 established separately.

382 A pass in one dimension is not promoted to another. In particular, finite
 383 output is not lifecycle safety, directional agreement is not a common Newton
 384 basin, and a single-element patch is not an engineering production benchmark.

385 6.2. Lifecycle and state-transaction tests

386 Lifecycle tests include equal-state replay, permuted rejected trials,
 387 clean versus forced-cutback histories, zero-increment and terminal-call
 388 insertion, output-on/output-off comparison, restart equivalence, and
 389 independent-process repetition. Physical-state comparison is distinct
 390 from diagnostic-writer comparison: a parallel output race can invalidate
 391 provenance without changing the single-element physical residual.

392 6.3. Registered geometry, states, directions, and thresholds

393 The elemental derivative tests use the unit square with nodes $(0, 0)$, $(1, 0)$,
 394 $(1, 1)$, and $(0, 1)$, four 2×2 Gauss points, unit thickness, and $\Delta t = 60$ s.
 395 Node-wise unknowns are (u_x, u_y, p_w, S_n) . The three affine trial states are
 396 $I0$: $u_x = 10^{-6}x$, $u_y = -5 \times 10^{-7}y$, $p_w = 1.20 \times 10^7 + 10^3x + 5 \times 10^2y$, and
 397 $S_n = 0.25 + 10^{-3}x + 5 \times 10^{-4}y$.

398 I1: $u_x = 10^{-5}x + 2 \times 10^{-6}y$, $u_y = -8 \times 10^{-6}y$, $p_w = 1.20 \times 10^7 + 2 \times 10^5x - 10^5y$,
399 and $S_n = 0.30 + 5 \times 10^{-3}x$.
400 I2: $u_x = 5 \times 10^{-4}x + 10^{-4}y$, $u_y = -3 \times 10^{-4}y + 5 \times 10^{-5}x$, $p_w = 1.20 \times 10^7 +$
401 $2.5 \times 10^5x - 1.5 \times 10^5y$, and $S_n = 0.25 + 0.12x + 0.04y$.
402 Characteristic variable scales are $X_u = 10^{-5}$ m, $X_p = 10^7$ Pa, and
403 $X_s = 10^{-1}$. The three 16-component mixed directions M1–M3 are normalized
404 only in these scaled coordinates and are tabulated without rounding in
405 Supplementary Table S4 and the machine-readable input. Central differences
406 retain all eleven registered perturbation levels,

$$h \in \{10^{-2}, 3 \times 10^{-3}, 10^{-3}, 3 \times 10^{-4}, 10^{-4}, 3 \times 10^{-5}, 10^{-5}, 3 \times 10^{-6}, 10^{-6}, 3 \times 10^{-7}, 10^{-7}\}. \quad (23)$$

407 The integrated nonzero-response gate is relative $L_2 \leq 10^{-5}$ with cosine
408 similarity at least 0.99999; registered null responses require scaled deriva-
409 tive norm, matrix-vector norm, and mismatch norms no greater than 10^{-10} .
410 Independent-process normalized difference is at most 10^{-13} . Branch-crossing
411 points remain in the tables but cannot support a smooth derivative claim.

412 The assembled patch is the $[0, 2] \times [0, 2]$ square subdivided into four unit
413 Q4 elements with nine nodes. The Newton benchmarks use manufactured
414 discrete loads $\text{RHS}(x_*)$, so the target root is known without changing the
415 residual. N1 uses state I1, direction M2, a primary scaled offset 0.75, and a
416 near-root offset 0.15; N2 uses I2, M3, 1.00, and 0.20. Local DOFs $\{1, 2, 3, 4, 6\}$
417 are fixed and $\{5, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16\}$ are free. The Newton gate
418 requires at least five residual states, three consecutive empirical orders $q \geq 1.8$,
419 a quadratic-constant ratio no greater than 10, final scaled residual $\leq 10^{-12}$,
420 and root error $\leq 10^{-10}$ without globalization or branch crossing. Complete
421 vectors and committed states are supplied in the reproducibility package.

422 6.4. Directional derivative protocol

423 For residual block R_a and variable block x_b , the registered central difference
424 is

$$D_h R_a[\mathbf{d}_b] = \frac{R_a(\mathbf{x} + h\mathbf{d}_b) - R_a(\mathbf{x} - h\mathbf{d}_b)}{2h}. \quad (24)$$

425 Each perturbation begins from the same committed state and discards its
426 candidate state. Three deterministic directions and 11 perturbation levels
427 from 10^{-2} to 10^{-7} are retained. Smooth nonzero responses use relative

428 error and cosine gates; null responses use scaled absolute gates. Branch-
 429 crossing records are retained but excluded from smooth central-difference
 430 claims. Independent-process repetition is part of the acceptance condition.
 431 The analytical oracle is implemented independently of the assembled matrix,
 432 reflecting the general principle that derivative generation and verification
 433 should not share the same computational path (Griewank and Walther, 2008).

434 Binary64 differences remain the primary smooth-core numerical test. A
 435 derivative oracle provides an independent analytical active-set path when hard
 436 clipping produces cancellation-limited differences, and binary128 arithmetic
 437 is used as supporting resolution evidence. Neither high precision nor an
 438 active-set oracle makes an exact switching point differentiable.

439 *6.5. Patch, Newton-order, and basin tests*

440 The assembled 2×2 Q4 patch includes rigid, affine mechanics, pressure,
 441 saturation, and mixed modes. Rollback sequences explicitly reject candidate
 442 state before replaying the accepted path. Newton-order benchmarks use
 443 manufactured discrete roots and fixed constraints, scales, initial offsets, and
 444 stopping rules. When those histories did not contain the required asymptotic
 445 segments, the failure was preserved and followed by resolution and local-map
 446 audits rather than relabelled as convergence.

447 Local-basin tests sample registered direction families, signs, and amplitudes
 448 in binary64 and binary128. Persistent entry requires the full fixed-band crite-
 449 rion, branch preservation, operator and decomposition closure, repeatability,
 450 and finite values. A direction without a registered entry is not automatically
 451 evidence of a Jacobian defect; projection, near-null alignment, manufactured-
 452 load mapping, nonlinear remainder, branch geometry, and resolution are
 453 audited as competing explanations.

454 *6.6. Output-provenance boundary*

455 Accepted-state output requires solver-managed record identity, complete
 456 element/Gauss-point coverage, finite fields, non-perturbation and same-build
 457 repetition. Diagnostic trial-call files do not satisfy that contract. A later
 458 diagnostic-only campaign established exact sharded writing for the registered
 459 single-thread case but quarantined the parallel trial-call CSV route rather
 460 than validating it. Subsequent record-route studies are summarized as in-
 461 frastructure limitations in Supplementary Table S6; they do not supply a
 462 full-domain result and are not required for the present lifecycle and Jacobian
 463 claims.

Table 4: Outcome matrix for the layered verification chain.

Evidence layer	Positive result retained	Limitation retained
State transaction	Identified CAP_REF path passes bounded replay and parity after solver-managed repair	No all-state or all-host-path claim
Six-block baseline	Residual is deterministic and individual block behavior is quantified	Matrix remains mixed quasi-Newton
Branchwise derivatives	Smooth-core six-block, nine-block and mixed directions pass	Strict active-branch record remains 36/39
Active-set supplement	Independent oracle and binary128 differences support three lower-clip derivatives	Exact switching is excluded
Patch and nonlinear tests	Available run1 patch/replay/conditioning gates pass	Complete patch/run2 and Newton-order gates are not established
Local basin	Persistent entry occurs in 23/26 combinations	No common radius; three directions remain mixed or unresolved

464 7. Verification evidence and failure-aware results

465 Positive, negative, and inconclusive outcomes are reported in the order of
466 the verification hierarchy. A result is promoted only within the layer exercised
467 by its registered test; later failures do not erase earlier local evidence, and
468 earlier passes do not substitute for benchmark qualification.

469 7.1. State-transaction counterexample and bounded repair

470 Static tracing and source-extracted standalone equal-declared-state replay
471 established the CAP_REF mechanism as a state-transaction defect: a rejected
472 trial changed a later residual at identical declared inputs in the unsafe
473 baseline. The component-wise replay records are distributed as machine-
474 readable evidence. After CAP_REF and its validity flag were moved to
475 SVARS(13:14), the same-state comparison was exact and all 300 shared
476 CAP_REF keys agreed in the bounded standalone parity experiment. This
477 demonstrates repair of the identified code path in the replay model; it is not,
478 by itself, a replayable Abaqus-host before/after experiment.

479 A repaired-build Abaqus patch supplied complementary host evidence.
480 The clean and forced-cutback paths each contained 19 accepted rows, and the
481 latter retained six rejected rows. Their maximum scaled accepted-state
482 wetting-pressure difference was 5.66×10^{-10} , below the predefined 10^{-8}
483 envelope, and the maximum reported nodal displacement difference was
484 4.65×10^{-40} m. Output-on/off comparisons and the registered restart path
485 were exact for the reported state records. Because the unsafe source, raw
486 solver .sta/.msg/.dat files, and an isomorphic before/after host run are not
487 all retained, this patch supports repaired-build bounded behavior but does
488 not independently reproduce the causal counterexample.

489 Physical-state evidence and diagnostic I/O evidence diverged under parallel
490 execution. The 14-thread physical result remained within its registered
491 envelope, but the trial-call diagnostic files showed duplicate or misplaced
492 headers. A later writer-isolation campaign passed exact single-thread sharding
493 and deterministic merge gates; the parallel trial-call route was ultimately
494 disabled at launcher level rather than shown to be non-perturbing. This
495 distinction leaves the repaired physical path supported by the bounded lifecycle
496 tests while excluding parallel diagnostic CSV and full-domain accepted-state
497 export from the present claim.

498 7.2. Rollback-safe quasi-Newton baseline as a controlled comparator

499 The six-block baseline audit evaluated six flow-related blocks over three
500 states, three directions, and 11 perturbation levels, producing 594 retained
501 records per independent run. The two processes were exactly reproducible.
502 K_{ps} was classified as numerically consistent over the registered domain.
503 K_{pu} and K_{su} confirmed the omission of the stress-dependent permeability
504 chain. K_{ss} remained a bounded approximation because storage-use, mobility,
505 capillary-curvature, and branch derivatives were incomplete. The original
506 K_{pp} mismatch was bounded by the matrix-only diagonal guard, whose charac-
507 teristic product was $10^{-12} \times 10^7 = 10^{-5}$. The uniform-pressure K_{sp} direction
508 was a discrete null response that the original relative metric could not classify
509 reliably.

510 These findings quantify, rather than disqualify, M1. Its residual and
511 repaired state transaction are the comparison baseline, but its mixed matrix
512 cannot be described as a complete physical Jacobian. The resulting omission
513 ledger determines the order of M2a corrections.

514 7.3. Branchwise physical-Jacobian evidence

515 The permeability-chain correction introduced a stateless permeability
516 value-and-derivative path and the resulting K_{pu} and K_{su} displacement chains.
517 Residual, closure, physical-state, replay, and non-target-block parity were ex-
518 act, while both corrected blocks satisfied the registered numerical-consistency
519 criteria. The mechanics-row correction then restored full-scale equivalent-
520 pressure mechanics coupling. The worst best-plateau relative errors for K_{uu} ,
521 K_{up} , and K_{us} were 7.30×10^{-13} , 2.45×10^{-14} , and 5.27×10^{-7} , respectively;
522 the affine mechanics patch and prior-stage regression controls passed.

523 The saturation-block correction completed the branchwise K_{ss} derivative
524 without changing the saturation residual. The worst smooth-interior plateau
525 error was 6.95×10^{-11} , the minimum registered branch-distance factor was
526 51.8, and the derivative-term sum closed to 1.11×10^{-16} . Uniform and affine
527 saturation subpatch errors were 2.34×10^{-12} and 2.50×10^{-11} , respectively.
528 Limiter-inactive, ratio-cap, beta-min, endpoint-clip, and stable final-clip states
529 were preserved as separate branches.

530 The stabilization-separation audit distinguished J_{phys} from the diagonal
531 numerical guard. Raw K_{pp} errors ranged from 2.15×10^{-4} to 2.16×10^{-3}
532 over the registered series, whereas the guard-subtracted physical block ranged
533 from 1.30×10^{-15} to 1.34×10^{-8} . The stabilization-error decomposition closed
534 to 8.80×10^{-18} . With a nullspace-aware absolute criterion for the uniform

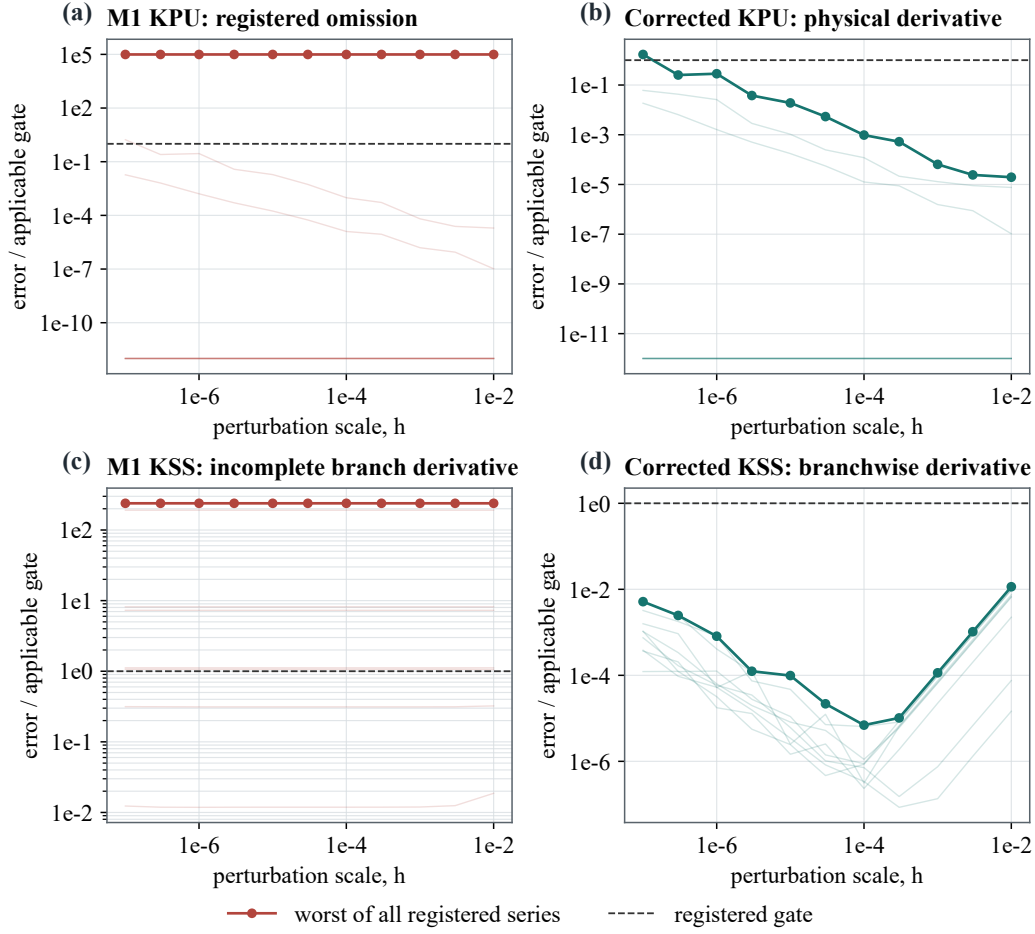


Figure 3: Representative complete perturbation curves distinguish the mixed quasi-Newton baseline from residual-preserving permeability-chain and saturation-block corrections. Individual panels do not establish universal nine-block consistency.

535 pressure direction, the worst non-null K_{sp} relative error was 3.44×10^{-9} . Thus
 536 the raw returned matrix and the physical residual derivative have different,
 537 explicitly reported classifications.

538 Integrated directional testing passed all six corrected flow-related blocks,
 539 all nine smooth-core blocks, and the smooth-core mixed full matrix. The
 540 preregistered strict record nevertheless remained 36 passing mixed series out
 541 of 39. The three FINAL_CLIP_LOW series had best binary64 relative errors
 542 of 5.03×10^{-5} , 1.52×10^{-5} , and 1.77×10^{-5} and were retained as inconclusive
 543 under the original finite-difference gate. The active-set supplement did not
 544 rewrite this result. Instead, an independent stable-active-set oracle produced
 545 relative errors of 2.58×10^{-16} , 8.74×10^{-17} , and 2.80×10^{-16} against $-J_{physd}$.
 546 The corresponding best binary128 finite-difference errors were 4.34×10^{-11} ,
 547 9.82×10^{-12} , and 9.33×10^{-16} . This supports the derivative on that stable
 548 active set and identifies binary64 cancellation as the limitation of those three
 549 central differences; it does not cover the exact hard-clip switching surface.

550 7.4. Assembled-patch evidence and nonlinear-convergence limits

551 The assembled patch and Newton-order stage used a 2×2 Q4 patch
 552 and exercised mechanics, pressure, saturation, and mixed affine modes. The
 553 available run1 mechanics, pressure, saturation, and mixed directional patch
 554 modes passed, as did the run1 equal-state, rejected-trial, forced-cutback,
 555 zero-increment, conditioning, and finite gates. The independent term-by-term
 556 global patch closure, M1/M2 parity, and second process were not run after the
 557 hard stop, so the stage is not promoted to a complete patch or repeatability
 558 result. The preregistered Newton-order gate was not met. N1 contained zero
 559 consecutive eligible empirical-order segments above 1.8, and N2 contained two
 560 rather than the required three. The registered Newton-order test therefore
 561 stopped because the required asymptotic sequence was unavailable; the patch
 562 and replay results were retained without promoting the Newton claim.

563 The resolution postmortem audited metrics, constraints, Taylor defects,
 564 cancellation, and a binary128 mirror. For N2, binary128 empirical orders
 565 of 1.933, 1.998, and 2.000 supported a local quadratic trend. For N1, only
 566 one transition remained eligible above the fixed 10^{-28} diagnostic floor, so the
 567 cause could not be uniquely classified. A favorable N2 sequence was not used
 568 to override the failed two-benchmark gate.

569 A frozen one-step-map audit then evaluated a frozen local one-step map
 570 without changing the target, initial directions, amplitudes, or operator. The
 571 registered root-error slopes were 0.374 for N1 and 0.318 for N2; residual

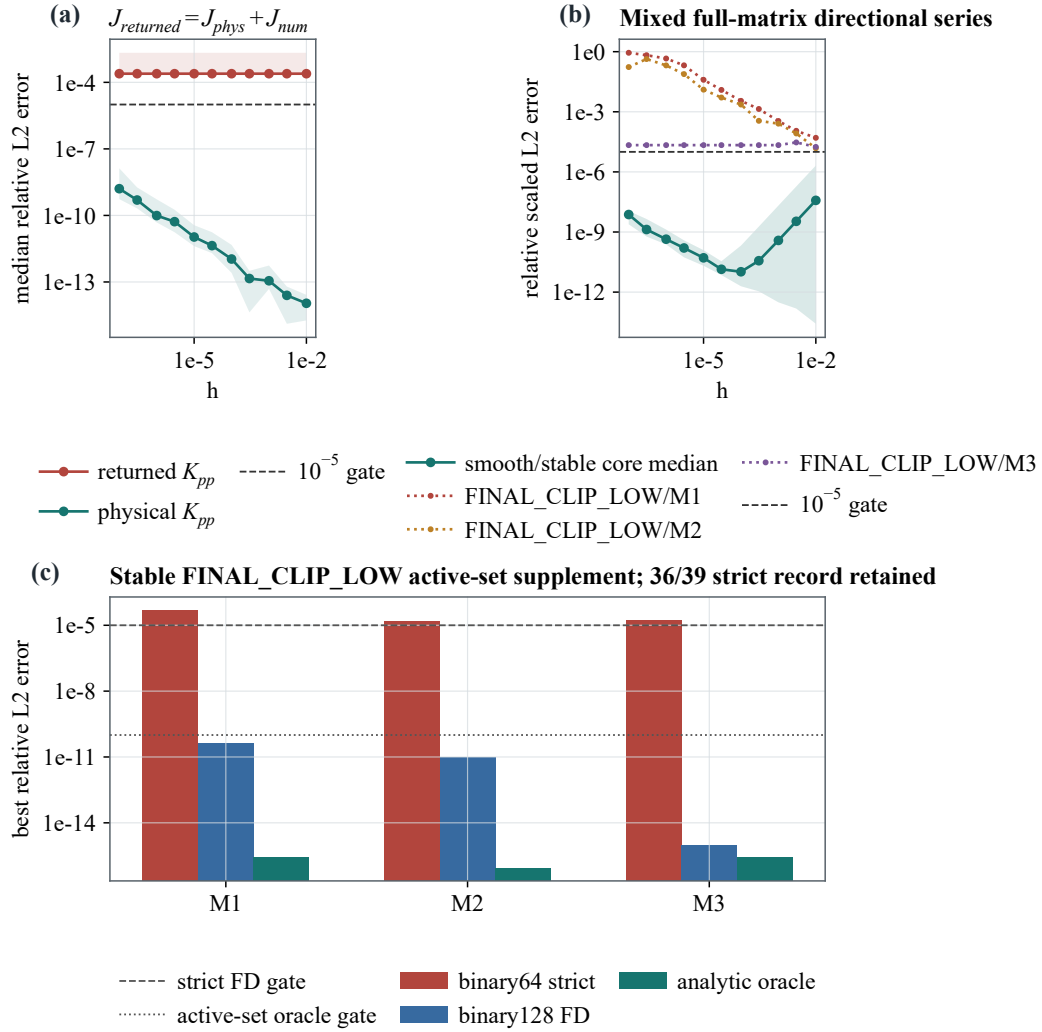


Figure 4: Numerical stabilization, smooth-core mixed directional evidence and the stable-active-set supplement are separate evidence layers. The oracle does not convert the original strict record into 39 of 39 central-difference passes.

572 slopes were 1.018 and 1.097. These results did not exhibit the preregistered
573 uniformly quadratic local map. The discrepancy was reproducible within that
574 audit, but no unique missing Jacobian term or single causal mechanism was
575 identified.

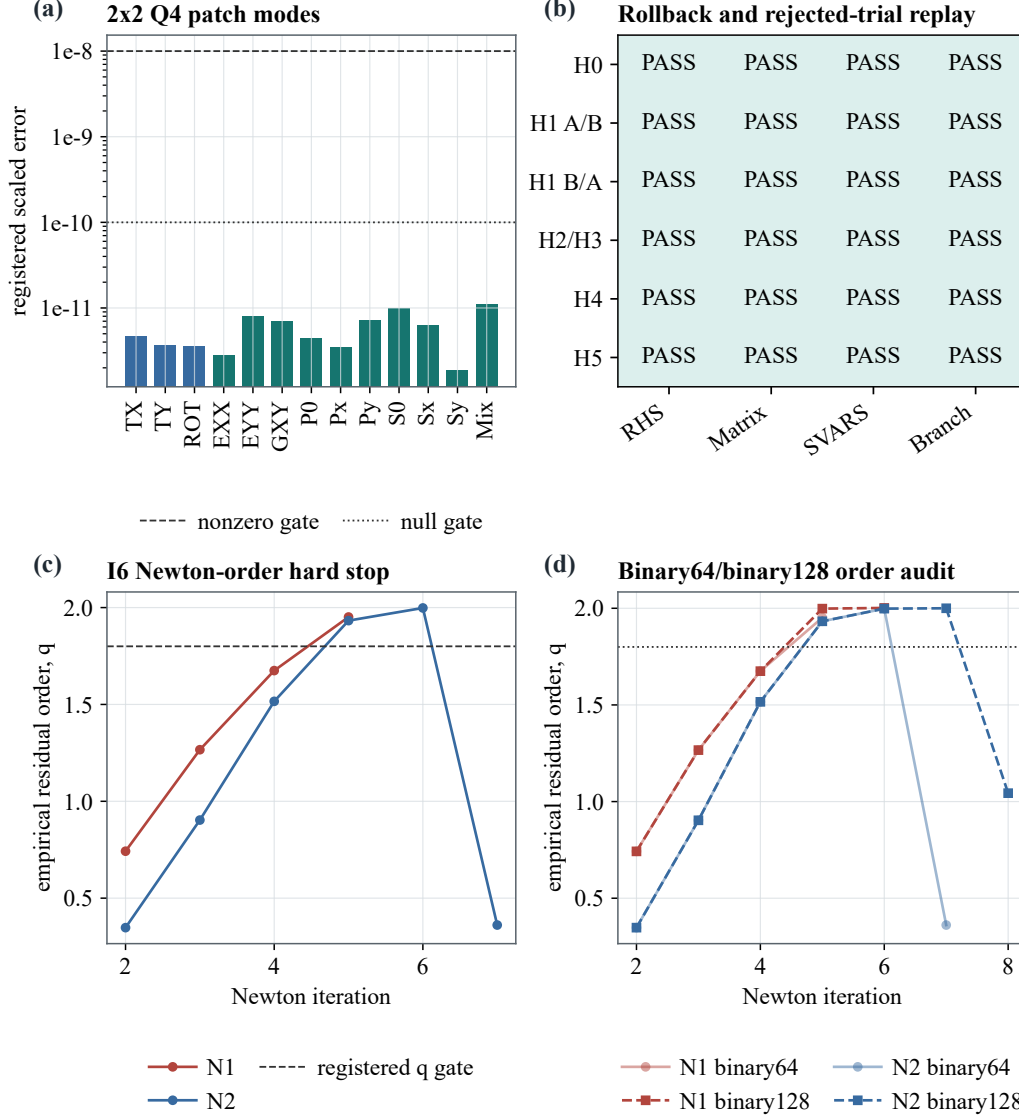


Figure 5: Available assembled directional-patch and replay gates passed, while the two-benchmark Newton-order and frozen local-map criteria remained unmet. High-precision N2 behavior is shown with, not instead of, the N1 limitation.

576 7.5. *Direction-dependent local basin and unresolved mechanisms*

577 The original local-basin run was blocked before classification because
578 mathematical unbounded margins were serialized as values exceeding the
579 finite-output contract and because two competing overlap norms had not been
580 uniquely frozen. Its map was not retrospectively classified. The corrected
581 local-basin rerun preregistered a bounded categorical margin representation,
582 one unit-floor overlap norm, and exact inheritance of the original legacy
583 directions without a second normalization. A complete independent rerun
584 passed source identity, overlap, operator, decomposition, replay, finite, and
585 repeatability gates.

586 The corrected map found persistent registered basin entry in 23 of 26
587 benchmark-direction combinations. No common registered radius was es-
588 tablished because N1/FIELD_U, N1/FIELD_P, and N2/FIELD_P had no
589 persistent entry. The observed basin was therefore direction dependent and
590 partial, not a common Newton basin.

591 The field-subspace audit froze those three target directions and seven
592 controls and audited state/residual projection, manufactured-load reduction,
593 scaled reduced operators, singular subspaces, one-step decomposition, branch
594 preservation, replay, finite values, and independent repetition. The maximum
595 registered directional operator defect was 1.38×10^{-13} . For N1/FIELD_
596 U, bottom-three singular-mode energy and response fractions were 0.471753
597 and 0.520872. For N1/FIELD_P, equal-component pressure alignment and
598 response fractions were 0.788200 and 0.762140. For N2/FIELD_P, both
599 equal-component fractions were 1.0, but the normalized operator response
600 was approximately 0.993 and the bottom-three singular-mode contribution
601 was negligible. None of the preregistered single-cause explanations reached
602 the 0.90 dominance gate. No single preregistered mechanism explained at
603 least 90% of any target response, so all three directions remain mixed or
604 unresolved. In particular, equal-pressure alignment is not by itself proof of a
605 gauge or null mode.

606 8. Discussion: verification hierarchy and evidence limits

607 The case study shows why user-element credibility cannot be reduced
608 to one convergence history or one final-field comparison. State ownership,
609 residual determinism, derivative consistency, assembly behavior, nonlinear
610 response, and engineering qualification are sequential but non-interchangeable
611 evidence layers.

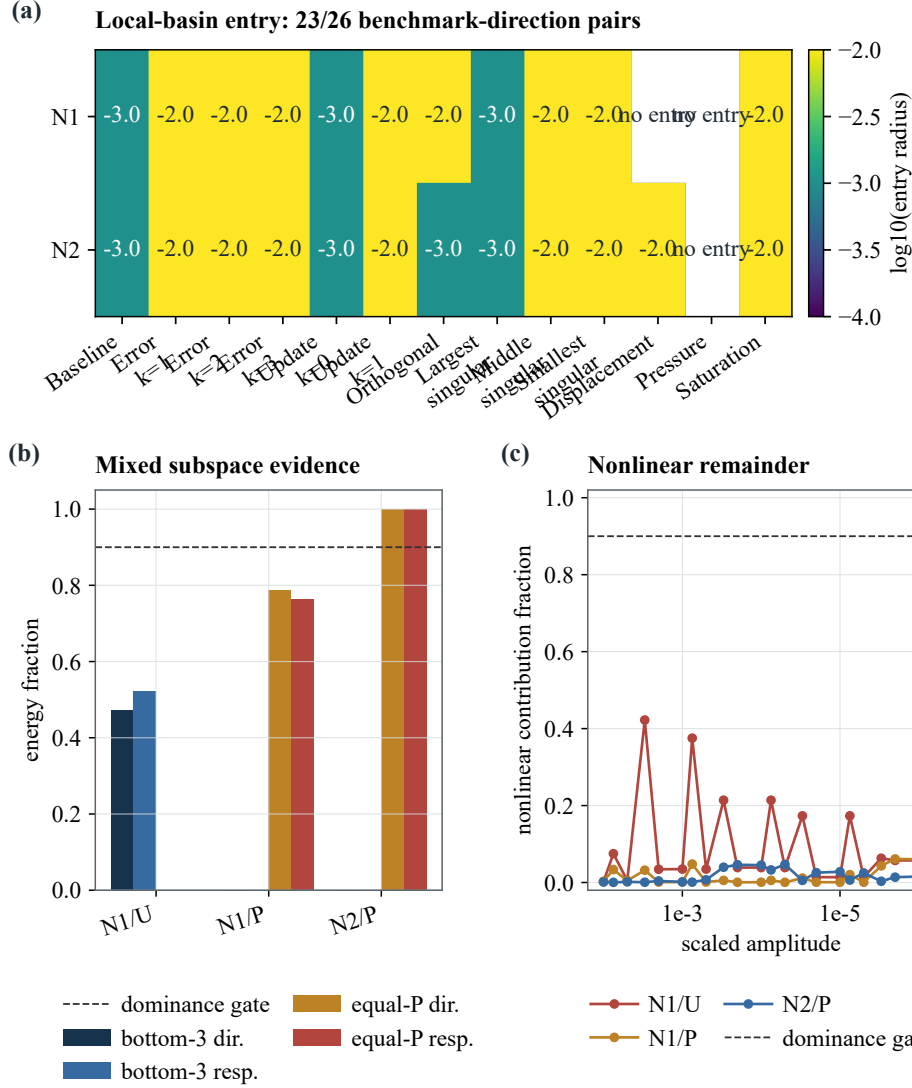


Figure 6: Persistent entry is direction dependent (23 of 26 combinations), and the three missing field-only directions remain mixed or unresolved after operator, projection and decomposition audits.

612 *8.1. Rollback consistency is not Jacobian consistency*

613 The CAP_REF defect and the M1 tangent omissions occupy different
614 logical layers. A hidden trial-updated cache makes the residual depend on
615 undeclared history. A quasi-Newton matrix approximates the derivative of
616 an otherwise deterministic residual. Repairing the first permits meaningful
617 differentiation tests; it does not make the second exact. Conversely, correcting
618 a tangent cannot restore a residual whose physical memory escaped host
619 rollback. Lifecycle and Jacobian claims must therefore be reported separately.

620 *8.2. Physical Jacobian and numerical matrix stabilization must be separated*

621 The K_{pp} audit illustrates why raw AMATRIX cannot be equated automati-
622 cally with a physical Jacobian. The unpaired diagonal guard predicts the raw
623 directional mismatch, but its identical unscaled coefficient across displacement,
624 pressure, and saturation diagonals is mixed-dimensional and unit dependent.
625 Subtracting it closes the registered physical derivative; it does not validate the
626 matrix received by the host, provide an inf-sup stabilization, or guarantee a
627 bounded saturation field. Low-order poromechanical formulations can require
628 explicit stability treatment and pressure-mode assessment (White and Borja,
629 2008). Likewise, conservative and bounded unsaturated-flow discretizations
630 require dedicated balance and maximum-principle evidence (Celia et al., 1990).
631 Those studies were not completed here.

632 *8.3. Directional consistency does not guarantee a common Newton basin*

633 The smooth-core nine-block and full-matrix directional tests are strong
634 local derivative evidence, yet the nonlinear map remained direction dependent.
635 Newton behavior also depends on the radius over which the local linearization
636 is informative, conditioning, mixed-field scaling, branch geometry, and higher-
637 order terms. The 23/26 entry result and the three unresolved field directions
638 show that accurate directional derivatives at registered states do not imply
639 one direction-independent attraction radius.

640 *8.4. Failed gates are verification evidence*

641 The Newton-order hard stop, inconclusive resolution audit, negative fixed-
642 band map, original output-contract failure and unresolved field-subspace
643 mechanisms are retained alongside positive evidence. They prevent selec-
644 tive reporting of the favorable N2 high-precision trend and rule out an
645 unconditional quadratic-convergence statement. Stopping further local-basin
646 expansion is a scope decision, not a declaration that the remaining nonlinear
647 behavior has been solved.

648 *8.5. Stable active-set evidence does not cover exact switching points*

649 The active-set supplement supports the lower hard-clip derivative only
650 where the active set is stable and separated from its switching surface. At
651 exact switching, a classical two-sided derivative need not exist. The active-
652 set oracle, binary128 supplement, and 36/39 strict binary64 record must
653 therefore remain visible together. Reporting the supplement as if it converted
654 the original record into 39 strict central-difference passes would erase the
655 intended distinction.

656 *8.6. Local evidence is not engineering qualification*

657 The single-element harness is valuable because it permits exhaustive state
658 control, branch isolation and independent residual differentiation. It does
659 not exercise full-domain assembly, boundary-condition provenance, global
660 conservation or long-horizon nonlinear continuation. The bounded writer
661 studies further show that diagnostic availability and physical-state validity
662 are separate questions. Full-domain validity and production readiness remain
663 unestablished.

664 *8.7. Discretization, conservation, and benchmark-qualification limits*

665 The element interpolates u_x , u_y , p_w , and S_n with the same bilinear Q4
666 basis. No inf-sup proof, pressure checkerboard study, mesh convergence,
667 time-step convergence, discrete maximum principle, or saturation-overshoot
668 study is available. The scalar matrix guard is not a substitute for these tests.
669 The registered RP and RS residuals also use different native balance units:
670 RP is an implemented wetting pressure-storage/volumetric balance, whereas
671 RS is a non-wetting mass balance. They cannot be summed as a two-phase
672 total-mass statement, and no accepted-increment boundary-flux ledger has
673 yet closed either full-domain balance.

674 The evidence is therefore code verification on controlled P1 states, not
675 solution verification or validation for a complex geotechnical application.
676 Before a stronger computational-method claim is resubmitted, the minimum
677 additional evidence is a documented benchmark with known or converged
678 behavior, mesh and time refinement, pressure-mode and saturation-bound au-
679 dits, separate native-balance closure, and a same-root comparison of $\mathbf{J}_{returned}$
680 and $\mathbf{J}_{phys}$ nonlinear behavior. Computational cost and robustness must also
681 be reported on that benchmark. The present manuscript retains these as
682 unmet gates rather than extrapolating from single-element derivatives.

683 8.8. *Implications for auditable UEL verification*

684 The evidence supports a staged verification order: state-transaction tests
685 should precede derivative tests; derivative evidence should precede assembled
686 nonlinear interpretation; and engineering claims should require an inde-
687 pendently qualified benchmark with discretization, balance, boundedness,
688 repeatability, and cost evidence. An iterative two-way fixed-stress scheme
689 remains a proposed extension rather than an inferred property of the present
690 monolithic element. Its trial, accepted and outer-iteration states would re-
691 quire separate versioning and a same-root comparison against the monolithic
692 formulation. Accepted-state full-domain output likewise remains a separate
693 protocol requiring solver-managed provenance, complete coverage, conserva-
694 tion observability and non-perturbation. Neither direction is presented as a
695 completed contribution of this paper.

Table 5: Hierarchical verification status and evidence boundaries.

Verification layer	Evidence established in this paper	Qualification boundary
State transaction	Six rollback conditions and the identified CAP_REF counterexample/repair	Safety of all persistent objects and all host paths is not established
Traceability	Bidirectional equation-to-code/state/Jacobian ledger	Thermodynamic consistency and material calibration are not established
Branch derivatives	Registered smooth and stable-active-set physical-Jacobian evidence	The strict record remains 36/39; exact-switch and raw-AMATRX completeness are not established
Assembly and nonlinear response	Available patch/replay evidence with retained Newton-order and basin limits	Complete patch repeatability, universal quadratic convergence, and a common basin are not established
Benchmark and deployment	Qualification requirements are defined explicitly	Mesh/time convergence, native conservation, full-domain validity, and production readiness are not established

696 9. Conclusions

697 This study established an auditable framework for separating state-
 698 transaction safety, residual determinism, derivative consistency, assembled
 699 behavior, and engineering qualification in a complex multiphysics UEL. Source
 700 tracing and standalone equal-state replay provided a falsifying case in which
 701 private trial-updated capillary-reference memory made a later residual depend
 702 on rejected history. Moving that reference and its validity flag into solver-
 703 managed state repaired the identified path in the replay tests; repaired-build
 704 Abaqus patch, cutback, and restart records provide complementary bounded
 705 host evidence rather than a complete unsafe-versus-repaired host replay.

706 The equation-to-code-state-Jacobian ledger separated residual determin-
 707 ism from derivative quality. The rollback-safe baseline retained known
 708 quasi-Newton approximations. Residual-preserving changes completed regis-
 709 tered branch derivatives for equivalent-pressure mechanics coupling, stress-
 710 dependent permeability, saturation storage and flux, and the physical part of
 711 the pressure block. Smooth-core block and mixed-direction tests support this
 712 branchwise physical-derivative candidate on the registered P1 states. The
 713 strict binary64 mixed record remains 36/39; the three lower-clip directions are
 714 supported only by a separately reported stable-active-set oracle and binary128
 715 evidence.

716 The nonlinear evidence places a stronger boundary on the claim. The
 717 preregistered Newton-order gate was not met, no uniform quadratic one-
 718 step map was observed, and persistent local-basin entry occurred for 23/26
 719 benchmark-direction combinations without a common radius. The remaining
 720 three field-only directions could not be assigned to one projection, near-null,
 721 load, branch, resolution, or Jacobian mechanism.

722 Accordingly, the principal result is an auditable pathway for testing
 723 rollback consistency, traceability, branchwise derivative correctness, and nu-
 724 merical reproducibility without promoting one evidence layer into another.
 725 For the frozen UEL, that pathway supports bounded state-ownership and
 726 selected branch-derivative claims over the controlled domain while preserv-
 727 ing the failed Newton-order gate and the partial basin result. Equal-order
 728 stability, separate native-balance conservation, mesh and time convergence,
 729 thermodynamic consistency, exact-switching derivatives, full-domain validity,
 730 and production readiness are not established. A fully coupled engineering
 731 benchmark satisfying convergence, balance, boundedness, repeatability, and
 732 cost criteria remains a separate qualification requirement.

733 **Data and code availability**

734 Code, data, figure scripts, selected host evidence, and manuscript-
735 supporting files are openly available at [https://github.com/yqma7980/ro](https://github.com/yqma7980/rollback-consistent-multiphase-uel-verification)
736 [llback-consistent-multiphase-uel-verification](https://github.com/yqma7980/rollback-consistent-multiphase-uel-verification) and are archived in
737 Zenodo at 10.5281/zenodo.21773371. Release v0.8.2 is self-contained for
738 rebuilding the six main and twelve supplementary figures from distributed
739 CSV inputs; its SHA-256 manifest covers the byte-preserved UEL snapshot,
740 integrated harness, frozen inputs and evidence, figure builders, and selected
741 repaired-build Abaqus patch material. Code is released under the MIT
742 License, while data, documentation, figures, and manuscript-supporting
743 material are released under Creative Commons Attribution 4.0 International.
744 The archive excludes licensed Abaqus software and does not contain the
745 complete unsafe-versus-repaired host replay; numerical regeneration requires
746 licensed Abaqus 2026 and Intel Fortran.

747 **Declaration of generative AI and AI-assisted technologies in the**
748 **writing process**

749 During preparation of this work, the authors used a generative AI-assisted
750 tool to support language editing, document structuring, consistency checks,
751 and the drafting and checking of post-processing, figure, packaging, and
752 LaTeX automation scripts. All numerical simulations were executed in the
753 authors' local computational environment; AI-assisted tools were not used
754 to generate, fabricate, or alter scientific data. The authors reviewed and
755 edited all assisted material, verified the source, evidence mappings, figures,
756 and references, and take full responsibility for the article.

757 **Declaration of competing interest**

758 The authors declare that they have no known competing financial interests
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764 **CRedit authorship contribution statement**

765 Weiji Sun: Conceptualization, Methodology, Software, Validation, Formal
766 analysis, Investigation, Data curation, Visualization, Writing – original draft,
767 Writing – review & editing, Supervision, Project administration, Funding
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770 Liang: Conceptualization, Methodology, Supervision, Writing – review &
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